

Learning Disentangled Representation for Cross-Modal Retrieval with Deep Mutual Information Estimation

Weikuo Guo

guoweikuo@mail.dlut.edu.cn
Dalian University of Technology
Dalian, Liaoning, China

Xiangwei Kong*

kongxiangwei@zju.edu.cn
Zhejiang University
Hangzhou, Zhejiang, China

Huaibo Huang

huaibo.huang@cripac.ia.ac.cn
University of Chinese Academy of Sciences
National Laboratory of Pattern Recognition, CASIA
Beijing, China

Ran He

rhe@nlpr.ia.ac.cn
University of Chinese Academy of Sciences
National Laboratory of Pattern Recognition, CASIA
Beijing, China

ABSTRACT

Cross-modal retrieval has become a hot research topic in recent years for its theoretical and practical significance. This paper proposes a new technique for learning such deep visual-semantic embedding that is more effective and interpretable for cross-modal retrieval. The proposed method employs a two-stage strategy to fulfill the task. In the first stage, deep mutual information estimation is incorporated into the objective to maximize the mutual information between the input data and its embedding. In the second stage, an expelling branch is added to the network to disentangle the modality-exclusive information from the learned representations. This helps to reduce the impact of modality-exclusive information to the common subspace representation as well as improve the interpretability of the learned feature. Extensive experiments on two large-scale benchmark datasets demonstrate that our method can learn better visual-semantic embedding and achieve state-of-the-art cross-modal retrieval results.

CCS CONCEPTS

• **Information systems** → **Multimedia and multimodal retrieval**.

KEYWORDS

cross-modal retrieval; mutual information estimation; disentangled representation learning

*Corresponding Author.

Permission to make digital or hard copies of all or part of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for components of this work owned by others than ACM must be honored. Abstracting with credit is permitted. To copy otherwise, or republish, to post on servers or to redistribute to lists, requires prior specific permission and/or a fee. Request permissions from permissions@acm.org.

MM '19, October 21–25, 2019, Nice, France
© 2019 Association for Computing Machinery.
ACM ISBN 978-1-4503-6889-6/19/10...\$15.00
<https://doi.org/10.1145/3343031.3351053>

ACM Reference Format:

Weikuo Guo, Huaibo Huang, Xiangwei Kong, and Ran He. 2019. Learning Disentangled Representation for Cross-Modal Retrieval with Deep Mutual Information Estimation. In *Proceedings of the 27th ACM International Conference on Multimedia (MM '19)*, October 21–25, 2019, Nice, France. ACM, New York, NY, USA, 9 pages. <https://doi.org/10.1145/3343031.3351053>

1 INTRODUCTION

The rapid growth of multimedia data on the internet gives rise to the strong demand for finding appropriate and effective methods to process the multi-modal data. This paper focuses on cross-modal information retrieval, which is one of the most fundamental research topics in the field of multimedia data processing.

The task of cross-modal information retrieval is to retrieve items of one modality (e.g. image) that are relevant to the given query of another modality (e.g. text). However, data from different modalities such as text, image, and video exhibit heterogeneous properties which makes it difficult to directly establish a connection among them. Such a difficulty is also called the "heterogeneous gap" and is the core problem to solve when dealing with data from different modalities. Thus, it plays a key role in the field of cross-modal retrieval to learn more effective visual-semantic embedding which can ensure the learned representation both intra-modality discriminative and inter-modality coherent.

For the past few years, with powerful learning ability and impressive performance, deep neural networks have been applied to many computer vision [17, 32, 48] and natural language processing [26, 43] tasks. In the area of cross-modal retrieval, many deep-learning-based methods have been proposed to bridge the heterogeneity gap between different modalities and achieve good performance [7, 11, 13, 39, 41]. The most common way to learn cross-modal visual-semantic embedding is to first encode each modality into their individual feature space. Then the features are mapped into a common subspace in which the relevance/similarity between different modality data can be quantitatively measured. The whole network is often optimized via a ranking loss based on

different constraints such as l_2 -norm or cosine distance to encourage the feature similarity of the same image-text pair to be greater than that of other negative pairs.

Although these deep-learning-based methods can learn the common subspace feature and achieve good performance, they still face some limitations and are insufficient to solve the adverse effects arise from the heterogeneous gap.

One limitation of the existing cross-modal retrieval methods is that they only try to constrain the learned common subspace representation without regarding how much useful information in the input image/text is transferred to it. One of the acknowledged properties of a "good" representation is that the learned representations should retain as much useful information of the input data as possible [1, 2, 16, 28]. The quantity of useful information in the learned representation is important for down stream tasks. To consummate this deficiency, a simple idea is to train a representation-learning network to maximize the mutual information (MI) between its inputs and outputs. However, mutual information is difficult to estimate especially in high-dimensional and continuous settings. Fortunately, the recent theoretical breakthrough has made it possible to effectively compute MI between high dimensional input/output pairs of deep neural networks [1, 16]. Early application attempts have been made to solve tasks like image generation [47], zero-shot learning [31] and reinforcement learning [34]. In this paper, we introduce deep mutual information estimation technique to cross-modal retrieval and design an mutual information maximizing module to simultaneously estimate and maximize the mutual information between input image or text data and the learned high-level representations.

Another limitation of the existing cross-modal retrieval methods is that the learned common subspace representation is highly entangled. The information contained in each modality would be different from each other. There is some information that is shared across different modalities which is the foundation to establish cross-modal connection. We call this kind of information the Modality-Shared Information (MSI). However, there exists information only belonging to each modality such as background information for image modality and cultural, history introductions for text modality. We call this kind of information the Modality-Exclusive Information (MEI). Most of the existing cross-modal retrieval methods embed different modality data into the common feature space in which these two kinds of information are highly entangled. However the MEI could be harmful for the retrieval performance, which requires the learned representations to expel the MEI and only preserve information that is shared across modalities. In order to obtain representations more suitable for cross-modal retrieval, in this paper, we propose a cross-modal disentangled representation learning module. With the proposed module the MSI and the MEI are factorized into different mutually independent factors in the latent representation respectively. The schematic diagram of cross-modal disentangled representation learning can be seen in Fig. 1.

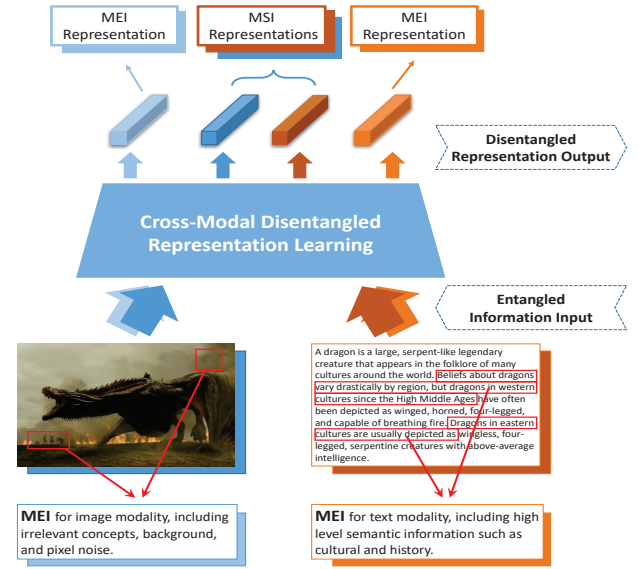


Figure 1: The Task of Cross-Modal Disentangled Representation Learning.

The above two modules are unified into a two-stage training architecture. During the first training process, the MI between the inputs and outputs of each subnet is estimated and maximized so that to make sure the learned representations uttermost retain information of the input data. During the second training process, the appended branches can further expel MEI and learn disentangled representations. Besides, to the best of our knowledge, this work is the first work that tries to solve the cross-modal retrieval problem with MI estimation and disentangled representation learning. The effectiveness of the MI constraint and two-stage training strategy is verified by the quantitative experimental results. The main contributions of this paper are summarized as follows.

- (1) An elaborate architecture is designed for cross-modal retrieval. With the two-stage training strategy and MI information estimation, the learned representation can well preserve the information which is useful for cross-modal retrieval.
- (2) We present the concept of cross-modal disentangled representation learning and design a modality-exclusive information expelling branch to diminish the side effects brought by MEI.
- (3) Experimental results on two public benchmarks show that the proposed method achieves competitive results compared with other state-of-the-art cross-modal retrieval approaches.

2 RELATED WORK

2.1 Cross-Modal Retrieval

The main goal of cross-modal retrieval is to bridge the heterogeneous gap between different modalities. Most of the existing

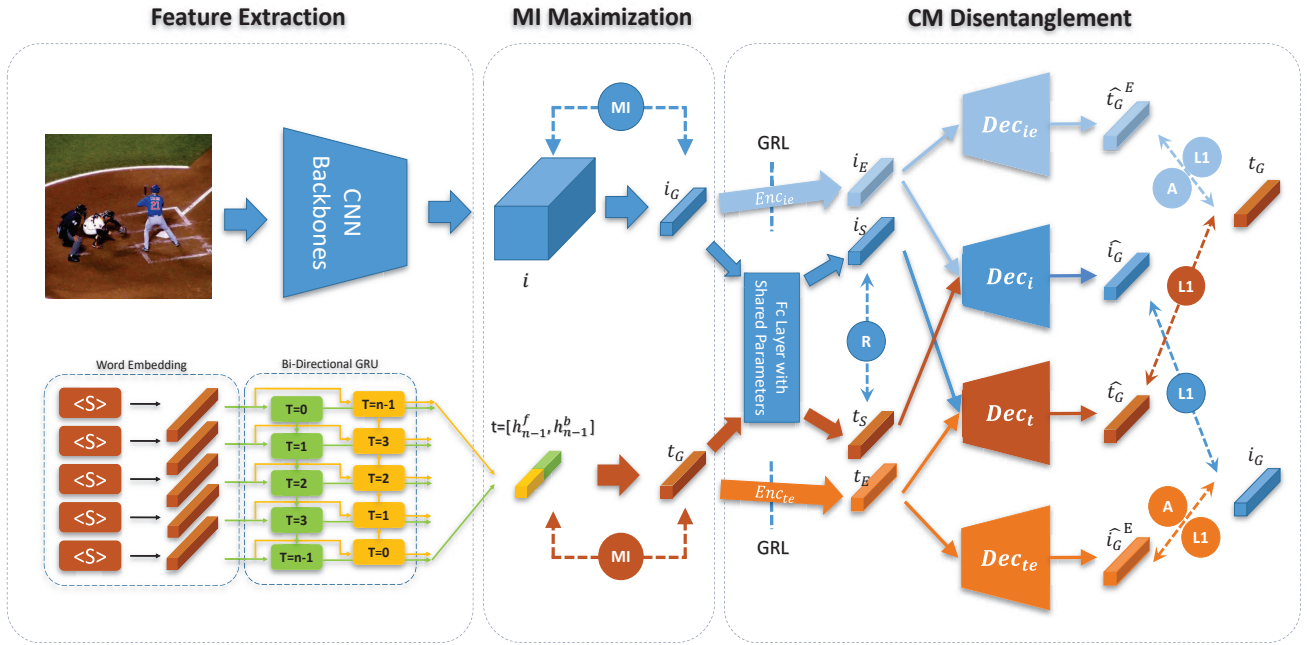


Figure 2: The Overall Architecture of the Proposed Method.

cross-modal retrieval methods try to solve this problem by learning common subspace representation. Based on the type of the learned representation, the existing cross-modal retrieval methods can be divided into two broad categories. Methods belong to the first category [6, 19] aim to project original data or features from different modalities into a common hamming subspace and get the binary representation, which are also called hashing methods. These methods achieve high computational efficiency because in the hamming space, the similarity between binary representations can be measured with hamming distance. However, such efficiency is at the cost of sacrificing the retrieval accuracy.

The method proposed in this paper falls into the other category called real-valued representation learning. In recent years, there are many researches have been done in this area. In [40], the authors introduce Generative Adversarial Network (GAN) to the task of cross-modal retrieval and propose to use a modality discriminator to adversely make distributions of the learned representation from different modality getting closer to each other. The method proposed in [11] tries to use raw data of one modality to generate data of the other modality with GAN and constrain the bottleneck representation with ranking-based loss. Huang et.al. [18] propose to use multi-modal context-modulated attention scheme to learn instance-aware saliency maps for image and sentence and then aggregate the local similarities into global similarity with LSTM network. From the perspective of practical application, Yu et.al. [45] propose a semi-supervised feature selection method for cross-modal retrieval. The proposed method adopts the graph-based constraint to predict labels

for unlabeled data, and keeps the geometric structure consistency between the label space and feature space so as to learn from unlabeled data. Zhan et.al. [46] propose to use a comprehensive distance-preserving autoencoder named CDPAE to solve the unsupervised cross-modal retrieval problem. With the proposed CDPAE, the correlations among different representations can be well explored.

2.2 Deep Mutual Information Estimation

Mutual information as a pivotal quantity across data science quantifies the dependence of two random variables. In contrast to correlation, mutual information captures non-linear statistical dependencies between variables and thus can act as a measure of true dependence[1]. The idea of maximizing MI between the input and output which is also called the infomax principle can be found in many traditional feature learning methods [2, 28]. However, mutual information is notoriously difficult to compute and estimate especially in high-dimensional and continuous scenarios therefore making it hard to adapt the infomax principle to the deep neural network.

Thanks to the recent theoretical breakthrough in deep MI estimation, the MI between the input and the output of deep neural network can be estimated and optimized. Belghazi et.al. [1] first propose a strongly consistent trainable MI estimator named Mutual Information Neural Estimator (MINE). The proposed MINE can estimate the MI of continuous variables via back-propagation and has been proved to be effective in many deep learning task settings. Following the pace of MINE, Hjelm et.al. [16] propose the Deep Infomax (DIM), which utilizes two different MI objectives that focus on local

or global structure to improve the suitability of representations for classification. Besides, there are many works extend the theoretical breakthrough to other application tasks such as images translating [25] and face recognition[38] and achieve good performance in their respective fields.

Following the above theoretical guidance, in this paper, we introduce MI estimation to the field of cross-modal retrieval and design an MI estimator to maximize the intra-modality global MI objective. Comparing with the existing cross-modal retrieval methods which only constraint common feature subspace, the proposed method enhance the retrieval performance by constraining the MI between input data and output representations thus can capture more useful information.

2.3 Disentangled Representation

The representation learned with deep learning is always highly entangled, which means all the information components of the input data are encoded into the high-dimensional feature space and affect the value of all the dimensions. People cannot exactly know specific information from it, which makes the learned representation hard to explicate. To solve this problem and make computing devices able to learn and reason in the same way as humans do, there have been many researches trying to learn disentangled representation with different approaches.

One commonly used architecture to learn disentangled representation is Generate Adversarial Network(GAN)[10]. Chen et.al. [3] propose InfoGAN to use mutual information objective to learn disentangled representations in an unsupervised manner. Liu et.al. [29] propose a multi-task adversarial network to learn disentangled face representation. By replacing factor in the learned latent representation with the target factor and constraining the generated image with the target one with adversarial loss, different information of interest in the input data are factorized into separate factors of the latent representation. Lu et.al. [30] use two encoders to encode content information and blur information into different factors and use KL-divergence to regularize the learned blur factor to contain little content information.

Another widely used architecture is variational auto-encoder (VAE) [23]. Higgins et.al. [15] introduce an adjustable hyperparameter β to balance the reconstruction and disentanglement quality and propose an unsupervised VAE-based framework named β -VAE. The drawback of β -VAE is that in order to obtain better disentangling result reconstruction performance must be sacrificed. To solve this problem Hyun-jik et.al. [21] introduce a new penalty that provides a better trade-off. Dupont [5] propose a new approach to improve β -VAE named JointVAE. The proposed method can simultaneously factorize continuous and discrete attributes of the input data in an unsupervised manner, which shows high flexibility when processing real-life data.

One of the most relevant research is [9] which mainly focuses on learning disentangled representations for cross-domain image translation. In [9], the authors try to factorize domain

shared information and domain specific information into mutually independent factors in the latent representation. Following their thinking, we introduce disentangled representation learning into cross-modal retrieval task. However, the heterogeneity of multi-modal data in our task makes it harder to learn disentangled modality-exclusive representation and we design a variety of strategies to help with the task.

3 METHODOLOGY

The overall framework of the proposed method can be seen in Fig. 3. The entire architecture can be roughly divided into three parts: multi-modal feature extracting(left), mutual information maximizing(middle) and cross-modal disentangled feature learning(right). In the following three subsections, we will introduce these three parts in detail respectively.

3.1 Multi-Modal Feature Extraction

The main goal of the first part is to learn multi-modal feature embedding. For image modality, each input image I is embedded to feature map I_{fm} . Depending on the chosen of CNN backbones, the dimension of I_{fm} is diversified. In our architecture, ResNet152 [14] and VGG19 [37] are used to extract image feature map, and the corresponding dimensions are $1024 * 7 * 7$ and $512 * 7 * 7$, respectively.

For text modality, the text input can be formulated as $T = (w_0, w_1, \dots, w_{n-1})$ where w_i denotes the one-hot code of the i th word in the text and n denotes the total text length. Following the common text feature embedding approach, we firstly use a word embedding matrix W_e to embed each word into a distributed embedding space, thus the i th word can be represented as $W_e w_i$. Then, we use a widely used sequence learning model: Bidirectional Gated Recurrent Unit Networks(Bi-GRU) to learn sentence representations. The Bi-GRU uses two hidden layers to learn from forward and backward information and as a result, can gather more useful information to the learned hidden representation. In this paper, we merge the last hidden representation of the forward GRU layer and the backward GRU layer and use it as the learned text representation which can be denoted as $t = [h_{n-1}^f, h_{n-1}^b]$. In consideration of simplicity, we formulate the first part as:

$$\begin{aligned} i &= I_{emb}(I, \theta_{i_{emb}}) \\ t &= T_{emb}(T, \theta_{t_{emb}}) \end{aligned} \quad (1)$$

where in our case $\theta_{i_{emb}}$ denotes the parameters of the image CNN backbone and $\theta_{t_{emb}}$ denotes the parameters of the word embedding matrix and the parameters of the Bi-GRU network.

3.2 Mutual Information Maximization

The goal of the second part is to learn two intra-modality feature learning subnets that satisfy the infomax principle. Without losing generality, the two subnet can be formulated as $i_G = I_{inf}(i, \theta_{i_{inf}})$ and $t_G = T_{inf}(t, \theta_{t_{inf}})$.

The core problem to solve in this part is how to design the MI estimator to simultaneously estimate and maximize

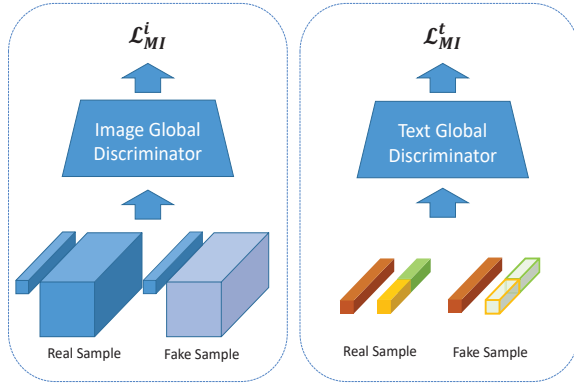


Figure 3: The Mutual Information Estimators of Two Modality.

the MI between the input(i/t) and the output(i_G/t_G) of each subnet. Following [1], we firstly rewrite the MI equation as the Kullback-Leibler (KL-) divergence between the joint distribution of two random variables and the product of their marginal, i.e. $I(k; k_G) = D_{KL}(\mathbb{P}_{kk_G} \parallel \mathbb{P}_k \otimes \mathbb{P}_{k_G})$, $k \in \{i, t\}$. The key technique to estimate the MI is to find an appropriate lower-bound for the KL-divergence. As introduced in [1, 16], the lower-bound can be obtained by representing the KL-divergence with Donsker-Varadhan representation [4] which has the following form:

$$D_{KL}(\mathbb{P} \parallel \mathbb{Q}) \geq \sup_{T \in \mathcal{F}} \mathbb{E}_{\mathbb{P}}[T] - \log(\mathbb{E}_{\mathbb{Q}}[e^T]) \quad (2)$$

where T can be concretized as a discriminator function modeled by deep neural network with parameters ω_k . So that the MI estimator $I_{\omega_k}(k; k_G)$ can be defined as:

$$I_{\omega_k}(k; k_G) := \mathbb{E}_{\mathbb{P}_{kk_G}}[T_{\omega_k}(k, k_G)] - \log(\mathbb{E}_{\mathbb{P}_k \otimes \mathbb{P}_{k_G}}[e^{T_{\omega_k}(k, k_G)}]) \quad (3)$$

During the training process, The expectations in Eq. 3 are estimated using empirical samples from the joint distribution and the product of their marginal and can be maximized with gradient back propagation. The powerful neural network insures that the designed MI estimator can approximate the mutual information with arbitrary accuracy.

In our proposed architecture, two intra-modality MI estimators are applied to each modality which can be denoted as T_i and T_t , the input of each estimator is the combination of the high-level feature i_G/t_G , and the original feature i/t . 'Fake' samples are drawn by combining the same high-level feature vector with original feature drew from another image/text which can be denoted as i'_G/t'_G . The 'real' samples and 'fake' samples pass through the discriminator T_i/T_t to get the score, and the binary cross entropy loss $\mathcal{L}_{MI} = \mathcal{L}_{MI}^i + \mathcal{L}_{MI}^t$ is calculated as the estimation of MI.

By adding MI estimation to the objective, the non-linear statistical dependencies between the low-level input feature and high-level output feature can maintain more useful information, which can benefit the following tasks.

3.3 Disentangled Representation Learning

With the help of MI estimation, the global feature learned with the previous two parts contains as much information of the input data as possible. However, not all of the information is beneficial for the task of cross-modal retrieval, because the information contained in each modality is not exactly the same. Roughly speaking, the information of each modality can be separated into two parts which are the Modality-Exclusive Information (MEI) and the Modality-Shared Information (MSI). The MEI represents the data redundancy or those factors of variation that are particular to each modality. Such information is harmful for the cross-modal retrieval task and should be restrained in the learned common subspace representation.

The main goal of the third part is two-fold. The task of disentangled representation learning requires the learned representation to be structured and clearly separated with respect to the kind of information. The task of cross-modal retrieval requires the learned common subspace representation to be intra-modality discriminative and inter-modality coherent. So that, in the third part, we design a disentangled representation learning network to encode the MSI and the MEI into different factors in the latent representation to expel MEI from the common subspace representation.

To learn modality-shared representation, we follow the common cross-modal representation learning approach to embed the features from different modality into a common subspace with a fully connected layer with shared parameters. The learned common subspace features are represented by i_S/t_S . Then we use a hinge-based triplet ranking loss to constrain the distance between features of ground truth caption-image pairs to be smaller than that of all other negative pairs. The ranking loss used in this paper is similar to other existing methods and has the following formulation.

$$\begin{aligned} \mathcal{L}_{Rank} = & \sum_{t'_S} \max(\lambda - s(t_S, i_S) + s(t'_S, i_S), 0) \\ & + \sum_{i'_S} \max(\lambda - s(t_S, i_S) + s(t_S, i'_S), 0) \end{aligned} \quad (4)$$

where i'_S and t'_S are the features of the negative samples and λ serves as a margin parameter. $s(t_S, i_S)$ represents the similarity measurement between two samples and in our architecture we use the order-violation penalty [39] which is defined as Eq. 5

$$s(t_S, i_S) = -\|\max(0, i_S - t_S)\|^2 \quad (5)$$

Only constraining common subspace feature with $\mathcal{L}_{Rank}$ can not ensure the completeness of the information in the learned representation. So that we refer to the process of auto encoder and reconstruct input data under the guidance of reconstruction loss to encourage all the information is encoded into the bottleneck representation. In our architecture, the information of the input data is disentangled into two factors which can be regard as the bottleneck representation, thus to reconstruct the input data, the designed decoder should

take these two factors as input. It is worth nothing that the learned modality-shared representation should contain the same information. To encourage the learned bottleneck representation to capture information that is shared across modalities and enhance the inter-modality coherence, we exchange the modality-shared representations when reconstructing. This process can be formulated as $\hat{i}_G = Dec_i(t_S, i_E)$ and $\hat{t}_G = Dec_t(i_S, t_E)$, and the l_1 norm reconstruction loss is minimized by:

$$\begin{aligned} \mathcal{L}_{Rec} = & \mathbb{E} [\|t_G - Dec_t(i_S, t_E)\|] \\ & + \mathbb{E} [\|i_G - Dec_i(t_S, i_E)\|] \end{aligned} \quad (6)$$

To learn modality-exclusive representation, we firstly encode the global feature i_G/t_G into the exclusive feature i_E/t_E with exclusive information encoder Enc_{ie}/Enc_{te} . The goal of disentangled representation learning requires the exclusive feature from one modality shouldn't contain information about the other modality, thus in ideal circumstances, we can not recover the other modality data with the exclusive feature. Following the above thinking, we decode the exclusive feature into $\hat{t}_G^E/\hat{i}_G^E$ and constrain the decoded feature and global feature from the other modality with adversarial loss. We add a Gradient Reversal Layer(GRL)[8] at the beginning of the decoder to invert the signs of gradient that backpropagated to the former layers. This strategy is at first proposed in [9] to learn domain-specific feature for cross-domain image translation. In their work, an adversarial loss is applied to the images generated by the domain-specific feature. However, in practice, we find it is hard to apply adversarial loss to our architecture for the reason that the two feature distributions are both shifting frequently in the early stage of training thus making the training frustratingly unstable.

We solve this problem from three aspects. Firstly, we adopt a more advanced WGAN-GP [12] loss as the adversarial loss. By measuring the distance between distributions with the Wasserstein distance and involving gradient penalty, WGAN-GP achieves more stable training and has been used in many other tasks. The generator and discriminator loss for image modality in our architecture can be defined as following and adversarial losses for text modality are similar.

$$\mathcal{L}_G^i = -\mathbb{E}_{t_G^E \sim \mathbb{P}_G} [D(\hat{t}_G^E)] \quad (7)$$

$$\begin{aligned} \mathcal{L}_D^i = & \mathbb{E}_{t_G^E \sim \mathbb{P}_G} [D(\hat{t}_G^E)] - \mathbb{E}_{t_G \sim \mathbb{P}_{data}} [D(t_G)] \\ & + \mathbb{E}_{t_G^E \sim \mathbb{P}_G} [\| \nabla_{t_G^E} D(\hat{t}_G^E) \|_2 - 1]^2, \end{aligned} \quad (8)$$

where $\mathbb{P}_{data}$ and $\mathbb{P}_G$ are the distributions of real data and generated fake data. The global adversarial loss for two modalities can be defined as $\mathcal{L}_{Adv} = \mathcal{L}_G^i + \mathcal{L}_G^t$.

Secondly, we add l_1 -norm loss to help the adversarial loss with constraining the distance between distributions, which can be defined as

$$\begin{aligned} \mathcal{L}_{Rec_E} = & \mathbb{E} [\|t_G - Dec_{te}(i_E)\|] \\ & + \mathbb{E} [\|i_G - Dec_{ie}(t_E)\|] \end{aligned} \quad (9)$$

Although this loss will not decline to a very low value because of the existence of the GRL layer, it can restrict the generated distribution to a limited range around the real one, thus making it easier for the adversarial loss to converge.

Thirdly, we also apply a two-stage training strategy to solve this problem. The overall loss function can be defined as:

$$\mathcal{L}_{all} = \mathcal{L}_{Rank} + \alpha \mathcal{L}_{MI} + \beta \mathcal{L}_{Rec} + \gamma \mathcal{L}_{Rec_E} + \delta \mathcal{L}_{Adv} \quad (10)$$

where α, β, γ and δ serve as tradeoff hyperparameters. During the first training stage, only $\mathcal{L}_{Rank}$, $\mathcal{L}_{MI}$ and $\mathcal{L}_{Rec}$ are used to obtain modality-shared representation. After the first stage converged $\mathcal{L}_{Rec_E}$ and $\mathcal{L}_{Adv}$ are activated to learn disentangled representations.

4 EXPERIMENTS

We conduct experiments on two widely-used datasets in the field of cross-modal retrieval i.e. MSCOCO dataset [27] and Flickr-30K dataset [44]. Comparisons with other state-of-the-art methods on these two datasets verify the effectiveness of our proposed method. Additional ablation study are presented in the later part of this section in order to dissect our method in detail.

4.1 Datasets and Experimental Setup

MSCOCO dataset and Flickr30K dataset are used to evaluate our method. MSCOCO is a widely used dataset for many tasks such as image caption and image segmentation. For cross-modal retrieval, we follow the common setting used in [7, 11, 20]. Under this setting, 113,287 samples are used for training and each sample contains one image and five corresponding text captions. Besides, 5000 samples are used for validation and 5000 samples are used for testing. In Flickr-30K dataset, 30,000 images are used for training. Following [7, 20], we use 1000 images for testing and 1000 images for validation.

To extract image feature map i_{fm} , we use two kinds of widely used CNN backbones in our experiments: ResNet152 and VGG19. We use the output of the average pooling layer for ResNet152 and the output of the last max pooling layer for VGG19 as the extracted feature map which has the dimension of 2048*7*7 and 512*7*7 respectively. To extract word original feature, we firstly encode every word into a one-hot code. The vocabulary size is 11,755. Then we embed every word into word embedding with a size of 300. The dimension of bi-directional GRU output is set to 1024. Besides, the dimension of global/shared/exclusive representations i_G/t_G , i_S/t_S , i_E/t_E and the dimension of reconstructed representations $\hat{i}_G/\hat{t}_G$, $\hat{i}_G^E/\hat{t}_G^E$ are all set to 1024.

The model is optimized with Adam[22] optimizer. We train our model with at most 30 epochs with a mini-batch size of 128. During the first 15 epochs only $\mathcal{L}_{Rank}$, $\mathcal{L}_{MI}$ and $\mathcal{L}_{Rec}$ are activated to learn the modality-shared representation and the learning rate is set to 0.0002. After the first stage converged, the modality-exclusive representation learning

branches are activated and the learning rate is decayed to 0.00002 for another 15 epochs.

The margin λ in the ranking loss Eq. 4 is set to 0.05 and the hyper-parameters α , β , γ and δ in Eq. 10 are set to 0.5, 0.5, 0.01, 0.5, respectively.

The evaluation indexes used in our experiments are the same with those in [7, 11]. A non-mainstream overall performance evaluation index named 'Sum' which is the summation of R@1 and R@10 scores of both i2t retrieval and t2i retrieval is extended to our experiments for fair comparison.

4.2 Experimental Results

Table 2: Cross-Modal Retrieval Comparison on Flickr30K Dataset

Model	Image-to-Text			Text-to-Image		
	R@1	R@10	Medr	R@1	R@10	Medr
UVS[24]	23.0	62.9	5.0	16.8	56.5	8.0
Embedding Net[41]	40.7	79.2	-	29.2	71.7	-
sm-LSTM[18]	42.5	81.5	2.0	30.2	72.3	2.0
DAN(ResNet)[35]	55.0	89.0	1.0	39.4	79.1	2.0
VSE++(ResNet)[7]	52.9	87.2	1.0	39.6	79.5	2.0
GXN(ResNet)[11]	56.8	89.6	1.0	41.5	80.1	2.0
Ours(ResNet)	57.7	89.2	1.0	42.9	79.3	2.0

4.2.1 Comparison with State-of-the-art Methods. We compare our method with other state-of-the-art methods on two datasets. For a fair comparison we use different CNN backbones to train our model and label it behind the name of methods.

Table 1 and Tabel 2 shows the cross-modal retrieval experimental results on MSCOCO dataset and Flickr30K dataset. From the results we have the following observations: 1) The proposed method outperforms other methods with a big margin on these two datasets. Experimental results on these two datasets which have very distinct properties can testify the effectiveness of our method. 2) On MSCOCO dataset, we conduct experiments with different CNN backbones with or without finetune. The performance of the proposed method outperforms other methods when the same CNN backbone setting is applied, which can prove our method has good applicability. 3) The best performance is achieved with ResNet152 with finetuning which shows that the choice of CNN backbones and finetuning operation can help with learning better image representations.

Note that comparing with our best competitor GXN [11] the performance of R@10 on Flickr30kK dataset doesn't improve very much but the R@1 value has a significant improvement. In real life retrieval scenario, we are more concern about the previous recalls, which means our method is more significant in practical applications. Besides, the GXN method used many subsidiary strategies and additional network to help with the model training which results in more complicated architecture, more parameters in the model and difficulty to apply to other architectures. Our model achieves competitive performance without adding too many

parameters to the model and can be easily applied to other architectures.

4.2.2 Ablation Study. In this subsection, we try to remove some network components or deactivate some items in the objective function and observe the effect on the model. We measure the performance by R@1 and R@10 score on MSCOCO dataset with 5000 test images. ResNet152 without finetuning is used as CNN backbones. The results can be seen in Table 3. The first row is our 'Baseline' result which is obtained with only using $\mathcal{L}_{Rank}$ as objective to train our model.

Table 3: Ablation Study on MSCOCO Dataset

Model	Image-to-Text			Text-to-Image		
	R@1	R@10	Medr	R@1	R@10	Medr
Baseline	38.4	76.0	3.0	26.9	68.0	4.0
Baseline+ $\mathcal{L}_{MI}$	39.9	79.8	2.0	28.1	69.7	4.0
Baseline+ $\mathcal{L}_{MI} + \mathcal{L}_{Rec}$	39.2	78.9	3.0	28.0	68.4	4.0
Stage1+ $\mathcal{L}_{Rec_E}$	39.4	80.0	3.0	29.7	71.3	4.0
Stage1+ $\mathcal{L}_{Adv}$	22.4	67.6	4.0	18.9	50.9	9.0
$\mathcal{L}_{all}$ without 2 Stage	27.4	77.6	6.0	22.9	61.8	7.0
Stage1+Stage2	40.2	80.5	2.0	29.9	71.9	3.0

Effect of Mutual Information Maximization. The first two rows are results obtained with or without $\mathcal{L}_{MI}$. From the results, we can see that by adding mutual information maximization between the input and output of a network, the performance of our model has a significant improvement. This reason should be attributed to that MI estimation can help to preserve more useful information in the learned representation and as a result, is beneficial to the retrieval task.

Effect of Disentangled Representation Learning. By comparing results in the 2nd row with the results in the 7th row, we can see the disentangled representation learning also improves the model's performance, which proves that the proposed method successfully learns disentangled information and expels the adverse impacts brought by modality-exclusive information.

Besides the overall assessment of the cross-modality disentanglement, we also discuss the effect of each branch in the disentangled representation learning module. By comparing the results in the 2nd row with the results in the 3rd row, we can see that only adding reconstruction loss $\mathcal{L}_{Rec}$ to the model does not improve the performance very much and even results in decline as the recall number increased. This may be because the bottleneck representations used for reconstruction should contain all the information in the input multi-modal data. As a result, the neural network would not only learn the information shared across different modalities but also learn the information that is useful to reconstruct the original representation.

Effect of $\mathcal{L}_{Rec_E}$ and Two-Stage Training. To learn the modality-exclusive representation, we use a two-stage training strategy and an additional $l1$ -norm loss $\mathcal{L}_{Rec_E}$ as an auxiliary. By comparing the results in the 4th, 5th, and 7th row, we can see that the performance drop significantly

Table 1: Cross-Modal Retrieval Comparison on MS-COCO Dataset

Model	Image-to-Text Retrieval			Text-to-Image Retrieval			Sum
	R@1	R@10	Med r	R@1	R@10	Med r	
	1k Test Images						
m-CNN[33]	42.8	84.1	2.0	32.6	82.8	3.0	242.3
HM-LSTM[36]	43.9	87.8	2.0	36.1	86.7	3.0	254.5
Oredr-embeddings[39]	46.7	88.9	2.0	38.9	85.9	2.0	260.4
DSPE+Fisher Vector[42]	50.1	89.2	-	39.6	86.9	-	265.8
sm-LSTM [18]	53.2	91.5	1.0	40.7	87.4	2.0	272.8
VSE++(Vgg19)[7]	51.9	90.4	1.0	39.5	85.6	2.0	267.4
VSE++(Vgg19, ft)[7]	57.2	93.3	1.0	45.9	89.1	2.0	285.5
GXN(Vgg19)[11]	51.4	91.4	1.0	39.1	86.7	2.0	268.6
Ours(Vgg19)	53.4	91.3	1.0	40.5	88.7	2.0	273.9
Ours(Vgg19,ft)	58.0	94.0	1.0	47.1	90.7	2.0	289.8
VSE++(ResNet152)[7]	58.3	93.3	1.0	43.6	87.8	2.0	283.0
VSE++(ResNet152, ft)[7]	64.7	95.9	1.0	52.0	92.0	1.0	304.6
GXN(ResNet152,ft)[11]	68.5	97.9	1.0	56.6	94.5	1.0	317.5
Ours(ResNet152)	59.4	95.1	1.0	46.5	90.8	2.0	291.8
Ours(ResNet152,ft)	68.9	97.0	1.0	57.3	94.4	1.0	317.6
	5k Test Images						
Oredr-embeddings[39]	23.3	65.0	5.0	18.0	57.6	7.0	163.9
VSE++(ResNet152)[7]	37.3	76.4	3.0	25.3	67.3	4.0	206.3
VSE++(ResNet152, ft)[7]	41.3	81.2	2.0	30.3	72.4	4.0	225.2
GXN(ResNet152, ft)[11]	42.0	84.7	2.0	31.7	74.6	3.0	233.0
Ours(ResNet152)	40.2	80.5	2.0	29.9	71.9	3.0	221.6
Ours(ResNet152,ft)	43.1	85.9	2.0	31.8	74.9	3.0	235.7

when we remove $\mathcal{L}_{Rec_E}$ and only use adversarial loss. From the results we can draw a conclusion that only when $\mathcal{L}_{Adv}$ and $\mathcal{L}_{Rec_E}$ work together can they assist each other and fulfill the task of learning modality-exclusive representation. By comparing the results in the 6th and 7th row, we can see when we use all the losses at the very beginning of the training process, the performance of the model significantly decrease. This result shows the two-stage training strategy can help with the task.

5 CONCLUSION

In this paper, we propose a new architecture to learn more effective and interpretable representations for cross-modal retrieval. Firstly, the proposed architecture introduce mutual information estimation to the task of cross-modal retrieval. By maximizing the mutual information between the input and output, our method can preserve more useful information for the visual-semantic embedding. Besides, we present the concept of cross-modal disentangled representation learning and design a cross-modal disentanglement module to solve it. With disentangled representation learning, our modal can expel the side effects brought by modality-exclusive information from the learned modality-shared features and benefit the cross-modal retrieval task. Extensive experiments on two widely used cross-modal datasets and a detailed analysis of the experimental results demonstrate the effectiveness of our method.

6 ACKNOWLEDGEMENT

We would like to thank anonymous reviewers for their helpful comments on the paper. This research was supported by the National Natural Science Foundation of China (NSFC) under Grant 61772111.

REFERENCES

- [1] Mohamed Ishmael Belghazi, Aristide Baratin, Sai Rajeshwar, Sherjil Ozair, Yoshua Bengio, Devon Hjelm, and Aaron Courville. 2018. Mutual Information Neural Estimation. In *ICML*. 530–539.
- [2] Anthony J Bell and Terrence J Sejnowski. 1995. An information-maximization approach to blind separation and blind deconvolution. *Neural computation* 7, 6 (1995), 1129–1159.
- [3] Xi Chen, Yan Duan, Rein Houthoofd, John Schulman, Ilya Sutskever, and Pieter Abbeel. 2016. Infogan: Interpretable representation learning by information maximizing generative adversarial nets. In *NIPS*. 2172–2180.
- [4] Monroe D Donsker and SR Srinivasa Varadhan. 1983. Asymptotic evaluation of certain Markov process expectations for large time. IV. *Communications on Pure and Applied Mathematics* 36, 2 (1983), 183–212.
- [5] Emilien Dupont. 2018. Learning disentangled joint continuous and discrete representations. In *NIPS*.
- [6] Venice Erin Liong, Jiwen Lu, Yap-Peng Tan, and Jie Zhou. 2017. Cross-Modal Deep Variational Hashing. In *ICCV*. 4097–4105.
- [7] Fartash Faghri, David J Fleet, Jamie Ryan Kiros, and Sanja Fidler. 2017. Vse++: Improving visual-semantic embeddings with hard negatives. *arXiv preprint arXiv:1707.05612* (2017).
- [8] Yaroslav Ganin and Victor Lempitsky. 2015. Unsupervised domain adaptation by backpropagation. In *ICML*. 1180–1189.
- [9] Abel Gonzalez-Garcia, Joost van de Weijer, and Yoshua Bengio. 2018. Image-to-image translation for cross-domain disentanglement. In *NIPS*. 710–720.
- [10] Ian Goodfellow, Jean Pouget-Abadie, Mehdi Mirza, Bing Xu, David Warde-Farley, Sherjil Ozair, Aaron Courville, and Yoshua

- Bengio. 2014. Generative adversarial nets. In *NIPS*. 2672–2680.
- [11] Jiuxiang Gu, Jianfei Cai, Shafiq R Joty, Li Niu, and Gang Wang. 2018. Look, imagine and match: Improving textual-visual cross-modal retrieval with generative models. In *CVPR*. 7181–7189.
 - [12] Ishaan Gulrajani, Faruk Ahmed, Martin Arjovsky, Vincent Dumoulin, and Aaron C Courville. 2017. Improved training of wasserstein gans. In *NIPS*. 5767–5777.
 - [13] Weikuo Guo, Liang Jian, Kong Xiangwei, Song Lingxiao, and He Ran. 2018. X-GACMN: An X-Shaped Generative Adversarial Cross-Modal Network with Hypersphere Embedding. In *ACCV*. 513–529.
 - [14] Kaiming He, Xiangyu Zhang, Shaoqing Ren, and Jian Sun. 2016. Deep residual learning for image recognition. In *CVPR*. 770–778.
 - [15] Irina Higgins, Loic Matthey, Arka Pal, Christopher Burgess, Xavier Glorot, Matthew Botvinick, Shakir Mohamed, and Alexander Lerchner. 2017. beta-vae: Learning basic visual concepts with a constrained variational framework. In *ICLR*, Vol. 2. 6.
 - [16] R Devon Hjelm, Alex Fedorov, Samuel Lavoie-Marchildon, Karan Grewal, Adam Trischler, and Yoshua Bengio. 2019. Learning deep representations by mutual information estimation and maximization. In *ICLR*.
 - [17] Ronghang Hu, Jacob Andreas, Marcus Rohrbach, Trevor Darrell, and Kate Saenko. 2017. Learning to Reason: End-To-End Module Networks for Visual Question Answering. In *ICCV*. 804–813.
 - [18] Yan Huang, Wei Wang, and Liang Wang. 2017. Instance-aware image and sentence matching with selective multimodal lstm. In *CVPR*. 2310–2318.
 - [19] Qing-Yuan Jiang and Wu-Jun Li. 2017. Deep Cross-Modal Hashing. In *CVPR*. 3232–3240.
 - [20] Andrej Karpathy and Li Fei-Fei. 2015. Deep visual-semantic alignments for generating image descriptions. In *CVPR*. 3128–3137.
 - [21] Hyunjik Kim and Andriy Mnih. 2018. Disentangling by factorising. *arXiv preprint arXiv:1802.05983* (2018).
 - [22] Diederik P Kingma and Jimmy Ba. 2015. Adam: A method for stochastic optimization. In *ICLR*.
 - [23] Diederik P Kingma and Max Welling. 2014. Auto-encoding variational bayes. In *ICLR*.
 - [24] Ryan Kiros, Ruslan Salakhutdinov, and Richard S Zemel. 2014. Unifying visual-semantic embeddings with multimodal neural language models. *arXiv preprint arXiv:1411.2539* (2014).
 - [25] Hsin-Ying Lee, Hung-Yu Tseng, Jia-Bin Huang, Maneesh Singh, and Ming-Hsuan Yang. 2018. Diverse Image-to-Image Translation via Disentangled Representations. In *ECCV*. 35–51.
 - [26] Lizhao Li, Guoyong Cai, and Nannan Chen. 2018. A Rumor Events Detection Method Based on Deep Bidirectional GRU Neural Network. In *ICIVC*. 755–759.
 - [27] Tsung-Yi Lin, Michael Maire, Serge Belongie, James Hays, Pietro Perona, Deva Ramanan, Piotr Dollár, and C Lawrence Zitnick. 2014. Microsoft coco: Common objects in context. In *ECCV*. 740–755.
 - [28] Ralph Linsker. 1988. Self-organization in a perceptual network. *Computer* 21, 3 (1988), 105–117.
 - [29] Yang Liu, Zhaowen Wang, Hailin Jin, and Ian Wassell. 2018. Multi-Task Adversarial Network for Disentangled Feature Learning. In *CVPR*. 3743–3751.
 - [30] Boyu Lu, Jun-Cheng Chen, and Rama Chellappa. 2019. Unsupervised Domain-Specific Deblurring via Disentangled Representations. In *CVPR*. 10225–10234.
 - [31] Chenguang Lu. 2019. The CM Algorithm for the Maximum Mutual Information Classifications of Unseen Instances. *arXiv preprint arXiv:1901.09902* (2019).
 - [32] Jiasen Lu, Caiming Xiong, Devi Parikh, and Richard Socher. 2017. Knowing When to Look: Adaptive Attention via a Visual Sentinel for Image Captioning. In *CVPR*. 375–383.
 - [33] Lin Ma, Zhengdong Lu, Lifeng Shang, and Hang Li. 2015. Multimodal convolutional neural networks for matching image and sentence. In *ICCV*. 2623–2631.
 - [34] Ofir Nachum, Shixiang Gu, Honglak Lee, and Sergey Levine. 2019. Near-Optimal Representation Learning for Hierarchical Reinforcement Learning. In *ICLR*.
 - [35] Hyeonseob Nam, Jung-Woo Ha, and Jeonghee Kim. 2017. Dual attention networks for multimodal reasoning and matching. In *CVPR*. 299–307.
 - [36] Zhenxing Niu, Mo Zhou, Le Wang, Xinbo Gao, and Gang Hua. 2017. Hierarchical multimodal lstm for dense visual-semantic embedding. In *ICCV*. 1881–1889.
 - [37] Karen Simonyan and Andrew Zisserman. 2014. Very deep convolutional networks for large-scale image recognition. In *ICLR*.
 - [38] Luan Tran, Xi Yin, and Xiaoming Liu. 2017. Disentangled Representation Learning GAN for Pose-Invariant Face Recognition. In *CVPR*. 1415–1424.
 - [39] Ivan Vendrov, Ryan Kiros, Sanja Fidler, and Raquel Urtasun. 2016. Order-embeddings of images and language. In *ICLR*.
 - [40] Bokun Wang, Yang Yang, Xing Xu, Alan Hanjalic, and Heng Tao Shen. 2017. Adversarial Cross-Modal Retrieval. In *ACM MM*. 154–162.
 - [41] Liwei Wang, Yin Li, Jing Huang, and Svetlana Lazebnik. 2018. Learning two-branch neural networks for image-text matching tasks. *TPAMI* 41, 2 (2018), 394–407.
 - [42] Liwei Wang, Yin Li, and Svetlana Lazebnik. 2016. Learning deep structure-preserving image-text embeddings. In *CVPR*. 5005–5013.
 - [43] Shuohang Wang, Sheng Zhang, Yelong Shen, Xiaodong Liu, Jingjing Liu, Jianfeng Gao, and Jing Jiang. 2019. Unsupervised Deep Structured Semantic Models for Commonsense Reasoning. In *NAACL*. 882–891.
 - [44] Peter Young, Alice Lai, Micah Hodosh, and Julia Hockenmaier. 2014. From image descriptions to visual denotations: New similarity metrics for semantic inference over event descriptions. *Transactions of the Association for Computational Linguistics* 2 (2014), 67–78.
 - [45] En Yu, Jiande Sun, Jing Li, Xiaojun Chang, Xianhua Han, and Alexander Hauptmann. 2018. Adaptive Semi-supervised Feature Selection for Cross-modal Retrieval. *IEEE TMM* 21, 5 (2018), 1276–1288.
 - [46] Yibing Zhan, Jun Yu, Zhou Yu, Rong Zhang, Dacheng Tao, and Qi Tian. 2018. Comprehensive Distance-Preserving Autoencoders for Cross-Modal Retrieval. In *ACM MM*. 1137–1145.
 - [47] Hao Zhu, Aihua Zheng, Huaibo Huang, and Ran He. 2018. High-Resolution Talking Face Generation via Mutual Information Approximation. *arXiv preprint arXiv:1812.06589* (2018).
 - [48] Lin Zhu, Yushi Chen, Pedram Ghamisi, and Jón Atli Benediktsson. 2018. Generative Adversarial Networks for Hyperspectral Image Classification. *IEEE Transactions on Geoscience and Remote Sensing* 56, 9 (2018), 5046–5063.